Homework_3_Question

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Logistics Regression

Homework to Environmental Statistics 3

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This weeks exercises will practice the content of the lecture and tutorials and *extend* them in different ways:

- using fitted curve to read off x for a given y (useful e.g. in LD50 value computation);
- 3D-plotting with added data points.

1 Exercise : Coronary heart disesase in San Francisco (12 points)

Coronary heart disease (CHD) is a common cause of death in the developed world. In this study, researchers wanted to find correlates of CHD across a range of study participants from around San Francisco. We will focus (medically inappropriately, but didactically importantly) on weight and cholesterol levels as "predictor" of CHD. Note that these are correlations, and hence in medicine they speak of "risk factors", not causal predictors.

1. Plot CHD incidence as functions of weight and cholesterol levels, separately. (**2 points**)
2. Analyse CHD incidence as function of weight, chostestrol and their interaction. Briefly say something about the model's quality and what the parameters suggest about the relationship between the two risk factors and CHD. (**3 points**)
3. Plot the interactions, and the probability of CHD as response (3D-like). Add the observed data points as "+". Remove extremes for weight and cholesterol to "zoom" into the part of the plot that is well-supported by measurements. `Re-plot` and add data as `.`(taking up less space). Choose a plot that shows a `scale bar` for colours. (**5 points**)
4. Interpret the relationship between weight, cholesterol and probability of CHD, i.e. put the statistical interaction into words. (**2 points**)

[2]:
```
# Import thew necessary libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import statsmodels.formula.api as smf
from statsmodels.formula.api import ols
import os
```

```
#os.chdir("/Users/phoebewells/Documents/environmental_statistics/week3")
```

[4]:
```
# load data
wcg = pd.read_csv("data/wcgs.csv")
wcgs = wcg.drop('year', axis=1)
wcgs.index = wcg['year']

print(wcgs.describe())
wcgs.head()
```

```
                age       height       weight          sdp          dbp  \
count  3154.000000  3154.000000  3154.000000  3154.000000  3154.000000
mean     46.278694    69.777743   169.953710   128.632847    82.015536
std       5.524045     2.528693    21.095762    15.117731     9.726880
min      39.000000    60.000000    78.000000    98.000000    58.000000
25%      42.000000    68.000000   155.000000   120.000000    76.000000
50%      45.000000    70.000000   170.000000   126.000000    80.000000
75%      50.000000    72.000000   182.000000   136.000000    86.000000
max      59.000000    78.000000   320.000000   230.000000   150.000000

              chol         cigs      timechd
count  3142.000000  3154.000000  3154.000000
mean    226.372374    11.600507  2683.859226
std      43.420426    14.517579   666.524135
min     103.000000     0.000000    18.000000
25%     197.250000     0.000000  2842.250000
50%     223.000000     0.000000  2942.000000
75%     253.000000    20.000000  3036.750000
max     645.000000    99.000000  3430.000000
```

[4]:

year	age	height	weight	sdp	dbp	chol	behave	cigs	dibep	chd	typechd	timechd	arcus
2001	49	73	150	110	76	225.0	A2	25	B	no	none	1664	absent
2002	42	70	160	154	84	177.0	A2	20	B	no	none	3071	present
2003	42	69	160	110	78	181.0	B3	0	A	no	none	3071	absent
2004	41	68	152	124	78	132.0	B4	20	A	no	none	3064	absent
2005	59	70	150	144	86	255.0	B3	20	A	yes	infdeath	1885	present

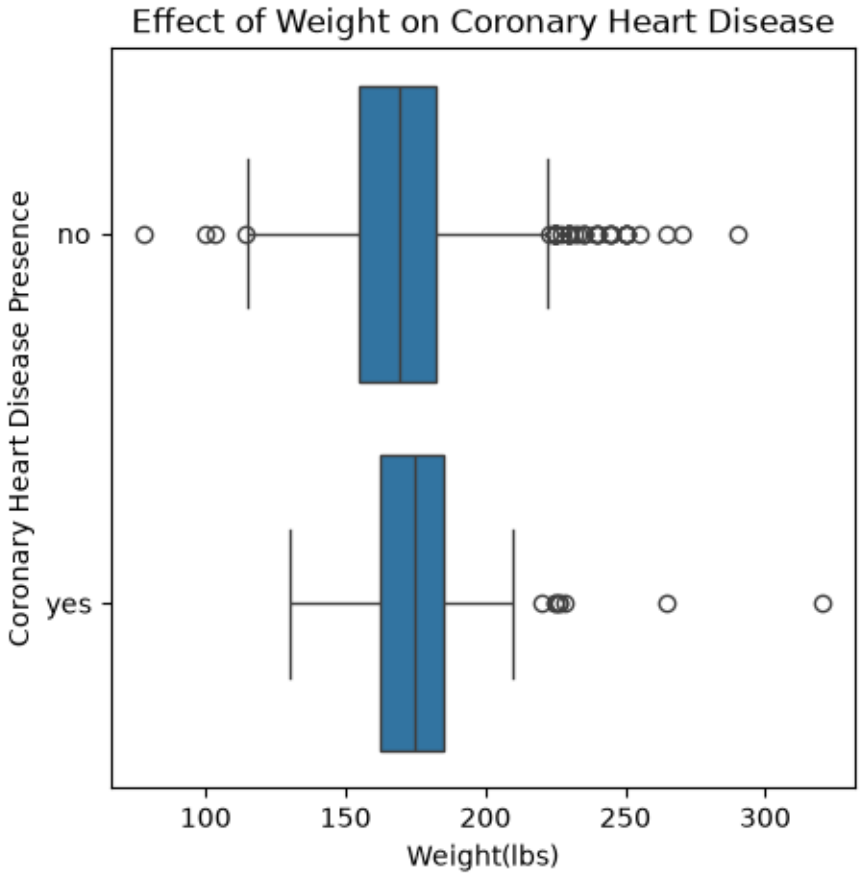
1.1 1. Plot

[5]:
```
# plot 1: CHD incidence as functions of weight

#convert chd to categorical
wcgs['chd']= wcgs['chd'].astype('category')

#create initial plot
plt.figure(figsize = (5,5))
#boxplot
sns.boxplot(data = wcgs, x='weight', y='chd')

#labels
plt.xlabel('Weight(lbs)')
plt.ylabel('Coronary Heart Disease Presence')
plt.title("Effect of Weight on Coronary Heart Disease")
plt.show()
```

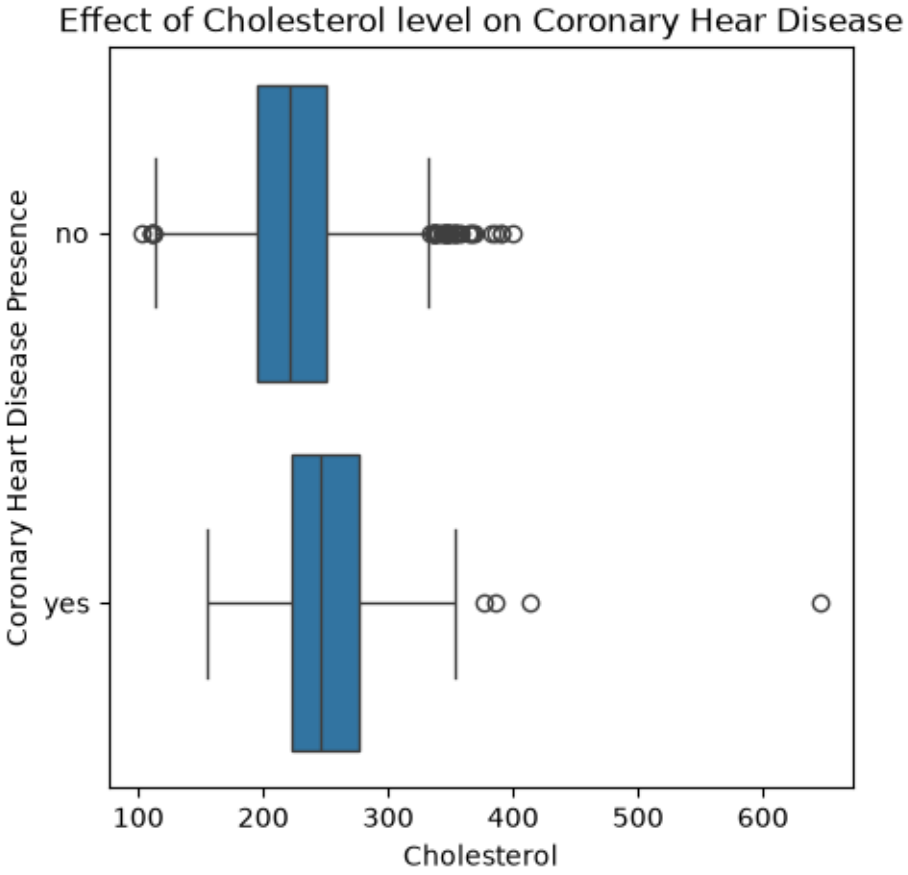


[6]:
```
# plot 2 : CHD incidence as functions of cholesterol levels
#initial plot
plt.figure(figsize=(5,5))

#boxplot
sns.boxplot(data = wcgs, x="chol", y="chd")

#labels
plt.xlabel("Cholesterol")
plt.ylabel("Coronary Heart Disease Presence")
plt.title("Effect of Cholesterol level on Coronary Hear Disease")

#make plot visible
plt.show()
```



1.2 2. Analyse

Analyse CHD incidence as function of weight, cholestrol and their interaction. Briefly say something about the model's quality and what the parameters suggest about the relationship between the two

risk factors and CHD.

[7]:
```
# model fit: Analyse CHD incidence as function of weight, cholestrol and their␣
 ↪interaction.
chd_model = smf.glm(data=wcgs, formula = 'chd~weight * chol', family= sm.
 ↪families.Binomial()).fit()
#show results
chd_model.summary()
```

[7]:

Dep. Variable:	['chd[no]', 'chd[yes]']	**No. Observations:**	3142
Model:	GLM	**Df Residuals:**	3138
Model Family:	Binomial	**Df Model:**	3
Link Function:	Logit	**Scale:**	1.0000
Method:	IRLS	**Log-Likelihood:**	-842.35
Date:	Mon, 09 Nov 2026	**Deviance:**	1684.7
Time:	13:05:53	**Pearson chi2:**	3.03e+03
No. Iterations:	6	**Pseudo R-squ. (CS):**	0.02963
Covariance Type:	nonrobust		

	coef	**std err**	**z**	**P> |z|**	**[0.025**	**0.975]**
Intercept	13.2134	2.580	5.121	0.000	8.157	18.270
weight	-0.0450	0.014	-3.109	0.002	-0.073	-0.017
chol	-0.0373	0.010	-3.588	0.000	-0.058	-0.017
weight:chol	0.0001	5.86e-05	2.420	0.016	2.7e-05	0.000

1.2.1 Analysis

The model shows significant p values for weight, cholesterol, and their interaction, implying that both variables have an impact on the presence of coronary heart disease in an individual. However, the Rsquared value of 0.029 tells us that this model only explains about 3 percent of the variance in the data. Therefore this suggests that a more effective model may need additional predictors.

1.3 3. Plot interaction

Plot the interactions, and the probability of CHD as response (3D-like). Add the observed data points as "+". Remove extremes for weight and cholesterol to "zoom" into the part of the plot that is well-supported by measurements. Re-plot and add data as "." (taking up less space). Choose a plot that shows a scale bar for colours.

[25]:
```
#3d plot including all data
#predict values using chd_model
weight_range = np.linspace(wcgs['weight'].min(), wcgs['weight'].max(), 50)
chol_range = np.linspace(wcgs['chol'].min(), wcgs['chol'].max(), 50)
weight_grid, chol_grid = np.meshgrid(weight_range, chol_range)
pred_data = pd.DataFrame({'weight': weight_grid.ravel(),'chol': chol_grid.
 ↪ravel()})
preds = chd_model.get_prediction(pred_data).summary_frame(alpha=0.05)
pred_grid = preds['mean'].values.reshape(weight_grid.shape)
```

```
#convert to 0,1
wcgs['chd_binary'] = wcgs['chd'].map({'no': 0, 'yes': 1})

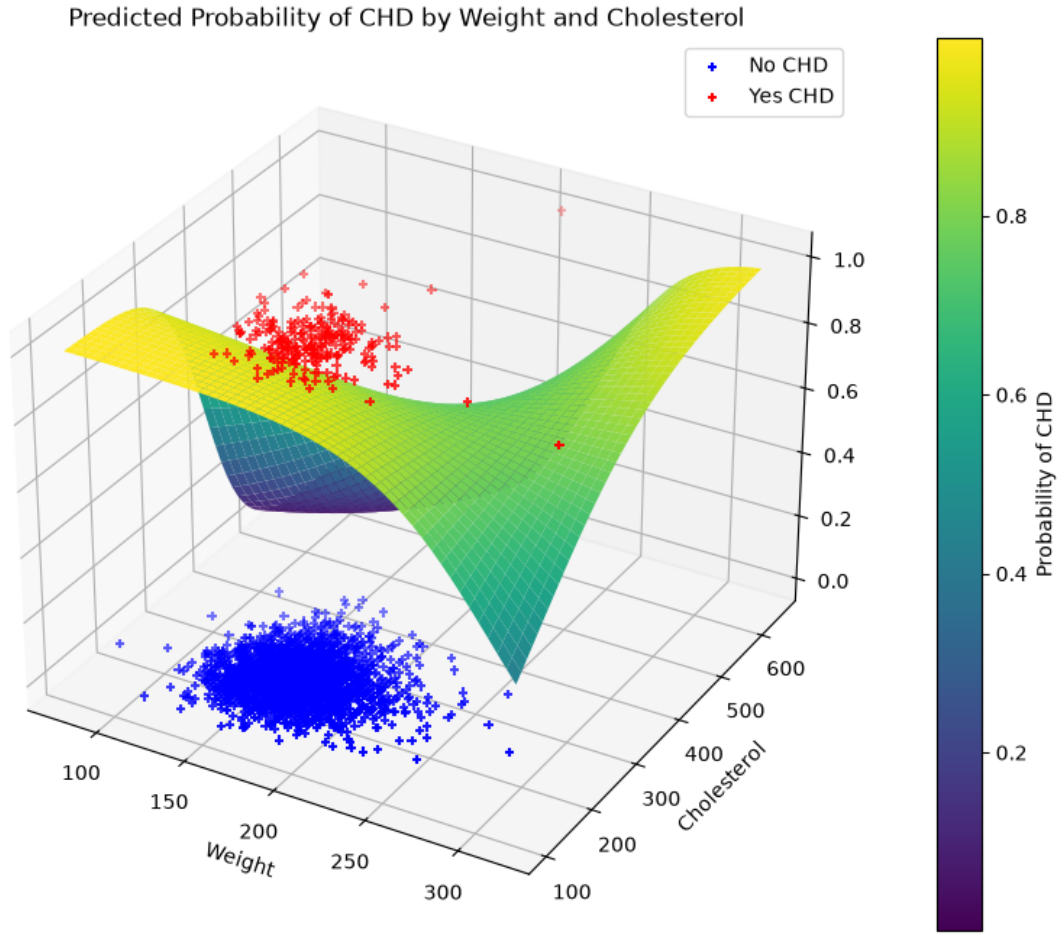
#blank plot and subplot
fig = plt.figure(figsize = (15,8))
ax = fig.add_subplot(111, projection = '3d')

#plot
surface = ax.plot_surface(
    weight_grid,
    chol_grid,
    pred_grid,
    cmap='viridis')

ax.scatter(
    wcgs[wcgs['chd_binary'] == 0]['weight'],
    wcgs[wcgs['chd_binary'] == 0]['chol'],
    wcgs[wcgs['chd_binary'] == 0]['chd_binary'],
    marker='+',
    color='blue',
    label='No CHD'
)

ax.scatter(
    wcgs[wcgs['chd_binary'] == 1]['weight'],
    wcgs[wcgs['chd_binary'] == 1]['chol'],
    wcgs[wcgs['chd_binary'] == 1]['chd_binary'],
    marker='+',
    color='red',
    label='Yes CHD'
)

ax.legend()
fig.colorbar(surface, ax=ax, label='Probability of CHD')
ax.set_xlabel('Weight')
ax.set_ylabel('Cholesterol')
ax.set_title('Predicted Probability of CHD by Weight and Cholesterol')
plt.show()
```



[27]:
```
#Zoomed in 3d plot
#remove top and bottom quantiles
wcgs_zoom = wcgs[
    (wcgs['weight'].between(wcgs['weight'].quantile(0.05),
                            wcgs['weight'].quantile(0.95))) &
    (wcgs['chol'].between(wcgs['chol'].quantile(0.05),
                          wcgs['chol'].quantile(0.95)))
].copy()

#predict values using chd_model
weight_range_zoom = np.linspace(wcgs_zoom['weight'].min(), wcgs_zoom['weight'].
 ↪max(), 50)
chol_range_zoom = np.linspace(wcgs_zoom['chol'].min(), wcgs_zoom['chol'].max(),␣
 ↪50)
weight_grid_zoom, chol_grid_zoom = np.meshgrid(weight_range_zoom,␣
 ↪chol_range_zoom)
```

```
pred_data_zoom = pd.DataFrame({'weight': weight_grid_zoom.ravel(),'chol':␣
 ↪chol_grid_zoom.ravel()})
preds_zoom = chd_model.get_prediction(pred_data_zoom).summary_frame(alpha=0.05)
pred_grid_zoom = preds_zoom['mean'].values.reshape(weight_grid_zoom.shape)

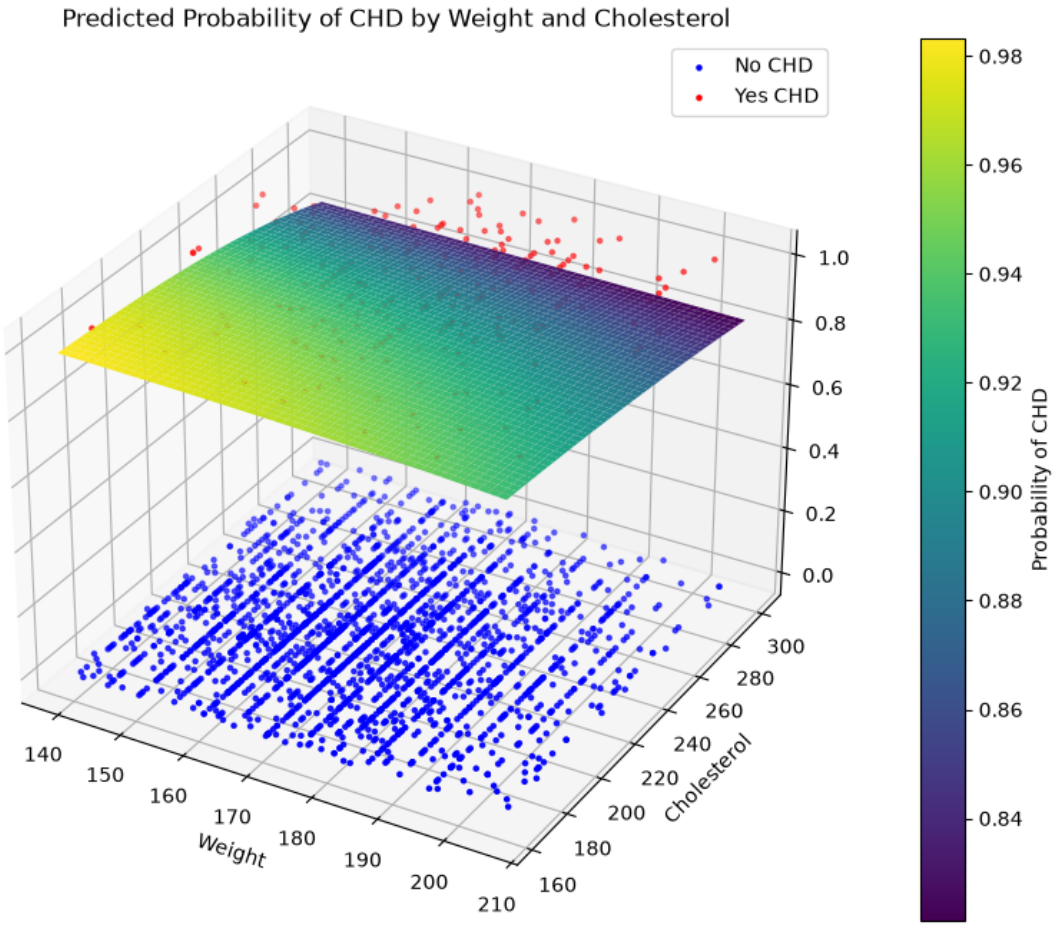
#blank plot and subplot
fig = plt.figure(figsize = (15,8))
ax = fig.add_subplot(111, projection = '3d')

#plot
surface = ax.plot_surface(
    weight_grid_zoom,
    chol_grid_zoom,
    pred_grid_zoom,
    cmap='viridis')

ax.scatter(
    wcgs_zoom[wcgs_zoom['chd_binary'] == 0]['weight'],
    wcgs_zoom[wcgs_zoom['chd_binary'] == 0]['chol'],
    wcgs_zoom[wcgs_zoom['chd_binary'] == 0]['chd_binary'],
    marker='.',
    color='blue',
    label='No CHD'
)

ax.scatter(
    wcgs_zoom[wcgs_zoom['chd_binary'] == 1]['weight'],
    wcgs_zoom[wcgs_zoom['chd_binary'] == 1]['chol'],
    wcgs_zoom[wcgs_zoom['chd_binary'] == 1]['chd_binary'],
    marker='.',
    color='red',
    label='Yes CHD'
)

ax.legend()
fig.colorbar(surface, ax=ax, label='Probability of CHD')
ax.set_xlabel('Weight')
ax.set_ylabel('Cholesterol')
ax.set_title('Predicted Probability of CHD by Weight and Cholesterol')
plt.show()
```



1.4 4. Interpret

Interpret the relationship between weight, cholesterol and probability of CHD, i.e. put the statistical interaction into words.

The model and plots show that weight and cholesterol have an impact on whether or not an individual is at higher risk of coronary heart disease, as the plotted probablity surface shows trends accross weight and cholesterol levels. In the first 3d plot we see a complex surface, with different trends showing across different weights and cholesterol values. This indicates that there is an interaction between the two predictions - reinforcing what we see in the model summary. This means that the effect of cholesterol on the risk of coronary heart disease changes depending on the weight of the individual. The zoomed-in plot shows less extreme differences in probability of coronary heart disease, indicated that more data may be needed at the extreme values for a more effective probability calculation at those levels.